Collecting Like Terms

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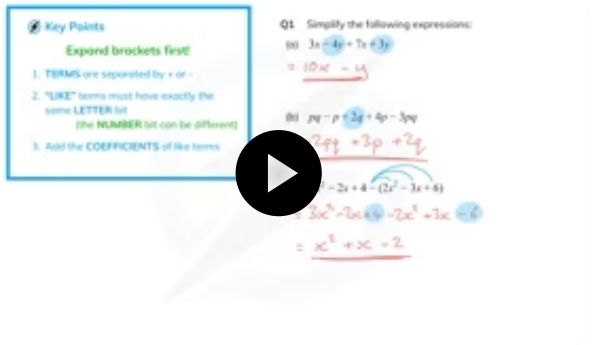

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Collecting like terms

What happens if there is more than one term?

- **Terms** can be **added** and **subtracted**
- The **numbers** in front of the letters are called **coefficients**
  - If the coefficient is 1 or -1, then it is conventional not to write the 1
    - The coefficient of x is 1
    - The coefficient of $-y$ s -1
- Each term has a positive or negative **sign** in front
  - In $2x - 3y$ the coefficient of x is 2 (positive) and the coefficient of $-3y$ is -3 (negative)
- **Subtractions** can be thought of as **adding** a **negative**
  - $2x - 3y$ is the same as $2x + (-3y)$
    - Just like 20 - 30 is the same as 20 + (-30)
- The **order** of two terms can be **swapped**, but the **signs** must **move** with their terms
  - $2x - 3y$ is the same as $-3y + 2x$
    - A **plus** is now needed in front of the $2x$
    - Just like 20 - 30 is the same as -30 + 20

What is a like term?

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

        - For example:
        - $2x$ and $3x$
        - $4x^2$ and $6x^2$
        - $5xy$ and $-7xy$
    - These are **not** like terms:
        - $2x$ and $3y$ (different letters)
        - $4x^2$ and $6x^4$ (different powers)
        - $5xy$ and $7xyz$ (different letters)
- Remember **multiplication** can be done in **any order**
    - xy and yx are **like** terms
        - So are $2xy$ and $3yx$

How do I collect like terms?

- **Collecting like terms** means simplifying by **adding** or **subtracting** the **coefficients**
    - $2x + 3x$ becomes $5x$
    - $4y - 10y$ becomes $-6y$
        - A negative sign is needed here
- If there are **different types** of like terms, **collect** them **separately**
    - For $2x + 4y + 5x - 3y$
        - Collecting the x's gives $2x + 5x = 7x$
        - Collecting the y's gives $4y - 3y = y$
        - The answer is $7x + y$

Examiner Tips and Tricks

Don't leave terms like $1x$ in your final answer in an exam - always simplify them to just x.

Worked Example

Simplify

$$8a - 5b - 6a + 4b$$

Answer:

Collect the a terms first

$$8a - 6a = 2a$$

Then collect the b terms
Don't forget the minus sign in front of the $5b$

$$-5b + 4b = -b$$

Add together the two answers

$$2a + -b$$

Simplify the signs

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